Higher-Order Differential Equations

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Today's Lecture

- Cauchy–Euler Equations
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Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Section 3.6**

Cauchy–Euler Equation

- A linear differential equation of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = g(x)$$

where the coefficients a_n, a_{n-1}, …, a_0 are constants, is known as a **Cauchy–Euler** equation.

- In Cauchy–Euler equations, the degree k = n, n − 1,…, 1, 0 of the monomial coefficients x^k matches the order k of differentiation $d^k y/dx^k$:

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots$$

❖ For example, for the nonhomogeneous second-order DE:

$$ax^2\frac{d^2y}{dx^2} + bx\frac{dy}{dx} + cy = g(x)$$

Get the general solutions of the associated homogeneous DE $ax^2y'' + bxy' + cy = 0$ using the Cauchy-Euler method.

Get the particular solution, y_p, by variation of parameters method.

❑ Method of solution

- For the associated homogeneous DE, we assume a solution of the form $y=x^m$ (where m is to be determined).
- After substituting x^m each term of a Cauchy–Euler equation becomes a polynomial in m times x^m.

e.g.: $ax^2y'' + bxy' + cy = 0 \longrightarrow am(m-1)x^m + bmx^m + cx^m = 0$

- So, $y=x^m$ is a solution of the differential equation whenever m is a solution of the auxiliary equation

$$am(m-1)x^m + bmx^m + cx^m = 0 \longrightarrow x^m[am(m-1) + bm + c] = 0$$

- There are three different cases to be considered, depending on whether the roots (m_1 and m_2) of this quadratic equation are real and distinct, real and equal, or complex.

  - **Case I:** m_1 and m_2 are real and distinct $\quad y = c_1x^{m_1} + c_2x^{m_2}$

  - **Case II:** m_1 and m_2 are real and equal $\quad y = c_1x^{m_1} + c_2x^{m_1}\ln x$

  - **Case III:** m_1 and m_2 are conjugate complex numbers $\quad y = c_1x^{m_1} + c_2x^{m_2}$

$$y = c_1x^{\alpha+i\beta} + c_2x^{\alpha-i\beta} \xrightarrow{\text{using Euler}} y = c_1x^{\alpha}\cos(\beta\ln x) + c_2x^{\alpha}\sin(\beta\ln x)$$


QUESTIONS

EXAMPLE 1

Solve $x^2y'' - 2xy' - 4y = 0$

EXAMPLE 2

Solve $4x^2y'' + 8xy' + y = 0$

- For higher-order equations, if m_1 is a root of multiplicity k, then it can be shown that

$$x^{m_1}, x^{m_1} \ln x\,, x^{m_1}(\ln x)^2, \ldots, x^{m_1}(\ln x)^{k-1}$$

are k linearly independent solutions. The general solution must then contain a linear combination of these k solutions.

EXAMPLE 3

Solve $4x^2y'' + 17y = 0$, $y(1)=-1, y'(1)=-0.5$

EXAMPLE 4

Solve $x^2y'' - 3xy' + 3y = 2x^4e^x$